

Gym & Fitness Centre Database

Advanced Database – Activity 1 · Group 5 · Data Dictionary and Team Contributions

Repository	advdb-activity1-Group5
Modelling tool	draw.io (diagrams.net), crow's foot notation
Normal form	All base relations in Third Normal Form (3NF)

1. Entity summary

ENTITY	DESCRIPTION	ATTRIBUTES
MEMBERSHIP_TYPE	Catalogue of membership plans offered by the gym (e.g. Basic, Standard, Premium).	6
MEMBER	A person registered with the gym who holds a membership plan.	14
PAYMENT_METHOD	Lookup list of accepted ways to pay.	3
PAYMENT	A single payment made by a member for a membership period.	10
STAFF	Any employee of the gym. Holds attributes shared by all staff; specialised into TRAINER or RECEPTIONIST.	7
TRAINER	A staff member who leads fitness class sessions.	4
RECEPTIONIST	A front-desk staff member who registers members and processes payments.	3
FITNESS_CLASS	A type of group class offered by the gym (the template for scheduled sessions).	7
FACILITY	A room or area of the gym where class sessions take place.	5
CLASS_SESSION	One scheduled occurrence of a fitness class, at a date/time, in a facility, led by a trainer.	8
ENROLLMENT	Resolves the M:N relationship between MEMBER and CLASS_SESSION; one row per member booked on a session.	4

Key column: PK = primary key, FK = foreign key, UQ = unique (alternate key), Derived = computed at query time and not stored as a column. Data types are generic SQL types for the logical model.

2. Data dictionary

MEMBERSHIP_TYPE

Catalogue of membership plans offered by the gym (e.g. Basic, Standard, Premium).

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
membership_type_id	PK	INT	Unique identifier of a membership plan.	Auto-increment; NOT NULL.
type_name		VARCHAR(30)	Name of the plan.	NOT NULL; UNIQUE.
monthly_fee		DECIMAL(8,2)	Current price charged per month for the plan.	NOT NULL; ≥ 0 .
duration_months		SMALLINT	Minimum contract length in months.	NOT NULL; between 1 and 36.
access_level		VARCHAR(20)	Level of facility access included.	IN ('Off-Peak', 'Full', 'All-Access').
description		VARCHAR(255)	Free-text summary of plan benefits.	Optional.

MEMBER

A person registered with the gym who holds a membership plan.

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
member_id	PK	INT	Unique identifier of a member.	Auto-increment; NOT NULL.
membership_type_id	FK	INT	Plan currently held by the member.	NOT NULL; references MEMBERSHIP_TYPE.
first_name		VARCHAR(50)	Member's given name.	NOT NULL; letters, spaces, hyphens only.
last_name		VARCHAR(50)	Member's family name.	NOT NULL.
email	UQ	VARCHAR(100)	Contact e-mail address; also used for login.	NOT NULL; UNIQUE; must match name@domain format.
phone		VARCHAR(20)	Contact phone number.	Digits, spaces, '+' and '-' only.
date_of_birth		DATE	Member's date of birth.	NOT NULL; member must be ≥ 16 years old at join_date.
age	Derived	INT	Member's current age in years.	Not stored; computed from date_of_birth and today's date.

gender		CHAR(1)	Self-reported gender.	Optional; IN ('F', 'M', 'X').
street_address		VARCHAR(100)	Street and house number.	Optional.
city		VARCHAR(50)	City of residence.	Optional.
postal_code		VARCHAR(10)	Postal / ZIP code.	Optional; alphanumeric.
join_date		DATE	Date the member first registered.	NOT NULL; DEFAULT current date; not in the future.
status		VARCHAR(10)	Current membership state.	NOT NULL; IN ('Active', 'Frozen', 'Expired', 'Cancelled').

PAYMENT_METHOD

Lookup list of accepted ways to pay.

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
payment_method_id	PK	INT	Unique identifier of a payment method.	Auto-increment; NOT NULL.
method_name	UQ	VARCHAR(30)	Name of the method (Cash, Credit Card, Debit Card, Bank Transfer, Mobile Money).	NOT NULL; UNIQUE.
is_active		BOOLEAN	Whether the method is currently accepted.	NOT NULL; DEFAULT TRUE.

PAYMENT

A single payment made by a member for a membership period.

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
payment_id	PK	INT	Unique identifier of a payment.	Auto-increment; NOT NULL.
member_id	FK	INT	Member who made the payment.	NOT NULL; references MEMBER.
payment_method_id	FK	INT	How the payment was made.	NOT NULL; references PAYMENT_METHOD.
processed_by	FK	INT	Receptionist who recorded the payment.	NOT NULL; references RECEPTIONIST(staff_id).
amount		DECIMAL(8,2)	Amount actually paid (historical value, kept even if the plan fee changes later).	NOT NULL; > 0.
payment_date		DATETIME	Date and time the payment was received.	NOT NULL; DEFAULT current timestamp.
period_start		DATE	First day covered by the payment.	NOT NULL.

period_end		DATE	Last day covered by the payment.	NOT NULL; period_end ≥ period_start.
transaction_ref	UQ	VARCHAR(40)	Receipt or bank/card transaction reference.	UNIQUE when present; required for non-cash methods.
payment_status		VARCHAR(10)	Outcome of the payment.	NOT NULL; IN ('Pending', 'Completed', 'Failed', 'Refunded').

STAFF (supertype)

Any employee of the gym. Holds attributes shared by all staff; specialised into TRAINER or RECEPTIONIST.

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
staff_id	PK	INT	Unique identifier of an employee.	Auto-increment; NOT NULL.
first_name		VARCHAR(50)	Employee's given name.	NOT NULL.
last_name		VARCHAR(50)	Employee's family name.	NOT NULL.
email	UQ	VARCHAR(100)	Work e-mail address.	NOT NULL; UNIQUE; valid e-mail format.
phone		VARCHAR(20)	Work phone number.	Digits, spaces, '+' and '-' only.
hire_date		DATE	Date the employee started.	NOT NULL; not in the future.
staff_type		CHAR(1)	Subtype discriminator.	NOT NULL; IN ('T', 'R'). Disjoint + total: every staff row has exactly one matching subtype row.

TRAINER (subtype of STAFF)

A staff member who leads fitness class sessions.

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
staff_id	PK, FK	INT	Identifier shared with the STAFF supertype.	NOT NULL; references STAFF; parent staff_type = 'T'.
specialty		VARCHAR(50)	Main area of expertise (e.g. Yoga, HIIT, Strength).	NOT NULL.
certification		VARCHAR(60)	Highest fitness qualification held.	NOT NULL.
hourly_rate		DECIMAL(6,2)	Pay rate per hour taught.	NOT NULL; > 0.

RECEPTIONIST (subtype of STAFF)

A front-desk staff member who registers members and processes payments.

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
staff_id	PK, FK	INT	Identifier shared with the STAFF supertype.	NOT NULL; references STAFF; parent staff_type = 'R'.
shift_pattern		VARCHAR(20)	Usual working shift.	NOT NULL; IN ('Morning', 'Afternoon', 'Evening', 'Rotating').
desk_extension		VARCHAR(6)	Internal phone extension at the front desk.	Optional; digits only.

FITNESS_CLASS

A type of group class offered by the gym (the template for scheduled sessions).

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
class_id	PK	INT	Unique identifier of a class type.	Auto-increment; NOT NULL.
class_name		VARCHAR(50)	Name of the class (e.g. Spin 45, Morning Yoga).	NOT NULL; UNIQUE.
description		VARCHAR(255)	What the class involves.	Optional.
category		VARCHAR(30)	Class family.	NOT NULL; IN ('Cardio', 'Strength', 'Mind-Body', 'Dance', 'Aquatic').
difficulty_level		VARCHAR(12)	Intended fitness level.	NOT NULL; IN ('Beginner', 'Intermediate', 'Advanced').
duration_minutes		SMALLINT	Length of every session of this class.	NOT NULL; between 15 and 180.
max_capacity		SMALLINT	Maximum participants per session.	NOT NULL; > 0; must not exceed the capacity of the facility used.

FACILITY

A room or area of the gym where class sessions take place.

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
facility_id	PK	INT	Unique identifier of a facility.	Auto-increment; NOT NULL.
facility_name	UQ	VARCHAR(50)	Name of the room/area (e.g. Studio A, Pool).	NOT NULL; UNIQUE.
room_type		VARCHAR(30)	Kind of space.	NOT NULL; IN ('Studio', 'Pool', 'Gym Floor', 'Outdoor').

capacity		SMALLINT	Safe maximum number of people.	NOT NULL; > 0.
floor_level		SMALLINT	Building floor where the facility is located.	Optional; between -2 and 10.

CLASS_SESSION (Schedule)

One scheduled occurrence of a fitness class, at a date/time, in a facility, led by a trainer.

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
session_id	PK	INT	Unique identifier of a scheduled session.	Auto-increment; NOT NULL.
class_id	FK	INT	Class being run.	NOT NULL; references FITNESS_CLASS.
trainer_id	FK	INT	Trainer leading the session.	NOT NULL; references TRAINER(staff_id); a trainer cannot lead two overlapping sessions.
facility_id	FK	INT	Room where the session is held.	NOT NULL; references FACILITY; no two overlapping sessions in the same facility.
session_date		DATE	Date of the session.	NOT NULL.
start_time		TIME	Start time of the session.	NOT NULL; within opening hours (05:00–22:00).
end_time	Derived	TIME	End time of the session.	Not stored; start_time + FITNESS_CLASS.duration_minutes.
available_spots	Derived	INT	Places still free in the session.	Not stored; max_capacity – COUNT(ENROLLMENT rows for the session).

ENROLLMENT (weak / associative)

Resolves the M:N relationship between MEMBER and CLASS_SESSION; one row per member booked on a session.

ATTRIBUTE	KEY	DATA TYPE	DEFINITION	VALIDATION RULES
member_id	PK, FK	INT	Member who booked the session.	NOT NULL; references MEMBER; member status must be 'Active'.
session_id	PK, FK	INT	Session that was booked.	NOT NULL; references CLASS_SESSION; bookings cannot exceed max_capacity.
enrollment_date		DATETIME	When the booking was made.	NOT NULL; DEFAULT current timestamp; must be before the session starts.

attendance_status

VARCHAR(10)

Whether the member attended.

NOT NULL; DEFAULT 'Booked'; IN ('Booked',
'Attended', 'No-Show', 'Cancelled').

3. Team contributions

TEAM MEMBER	ROLE	SPECIFIC TASKS
[Team member 1 name]	Project Lead / Business Analyst	Coordinated the group, gathered requirements for the gym scenario, wrote the business rules, created the GitHub repository.
[Team member 2 name]	Data Modeller	Identified entities and attributes, defined primary/foreign keys, relationships, cardinalities and participation.
[Team member 3 name]	EERD Designer	Built the EERD in draw.io (specialization, weak entity, derived attributes) and exported the diagram image.
[Team member 4 name]	Normalization Analyst	Checked every relation against 1NF, 2NF and 3NF and documented the key normalization decisions.
[Team member 5 name]	Documentation Lead	Wrote the data dictionary and validation rules, the README, and produced this PDF.

All team members are listed, including those whose contributions were limited.